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Towards Scarless Support Removal in Vat Photopolymerization Additive Manufacturing for Mass Manufacturing of Precision Components

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Abstract

This paper explores optimization techniques for support structures in Vat Photopolymerization Additive Manufacturing (VPP AM), addressing challenges posed by overhangs and inclined edges less than 45°. This research aims to overcome manufacturing optimization limitations in situations where the support structure is needed, by generating a benchmark methodology for further automated industrial mass production. Success criteria include ensuring part attachment to the support structure, minimal scarring after support removal, and high fabrication fidelity. Experiments are focused on a worst-case scenario involving a simple bar geometry positioned at 0° to the surface with support attached underneath. This setup facilitated precise control of process parameters and mask optimization per layer. Tests were conducted using the open-architecture ultra-high resolution VPP system, developed by the AM group at Technical University of Denmark (DTU), and commercial photopolymer resin. The study investigated the impact of Support-Free Factor (SFF) value, and Process Parameters (PP) for different support generation techniques. Variations in UV light intensity at the point of contact between the support tip and the part revealed correlations with final surface quality. Notably, a decrease in Z-axis dimensional error from 20% to 6.9% was observed during the study. The minimum required SFF for the VPP AM system was identified being equal to 65.19, corresponding to 1.51% of the supported area of the part. Furthermore, the study found that increased support structure and early part removal before post-processing resulted in finished part dimensions closest to the CAD model. Additionally, under-curing within the part structure improved the surface finish and achieved thicknesses closer to nominal values. These findings contribute to the advancement of VPP AM by providing insights into support structure optimization and process parameter adjustments for enhanced part quality and manufacturing efficiency.

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Keywords: Additive Manufacturing Systems; Vat Photopolymerization; Support Removal; Mass 3D Manufacturing; Rapid Prototyping

1. Introduction

Additive Manufacturing (AM) has revolutionized the production landscape by offering unprecedented flexibility and efficiency in creating intricate geometries. In the era of Industry 4.0, characterized by intelligent networks of machines, people, and products powered by data, AM stands at the forefront of transformative technological advancements. Among various AM techniques, Vat Photopolymerization (VPP), has garnered significant attention for its ability to fabricate high-resolution parts with exceptional surface finish [1]. The utilization of VPP machines introduces digitalization and automation to revolutionize production processes, reducing tooling costs, increasing design flexibility, and enabling mass customization. Using the principle of mass customization the cost of producing one unit can be less than that of traditional mass production methods [2].

For the past decade, companies involved in 3D manufacturing have been moving towards mass production and automation of the process chain (Carbon3D, Stratasys,

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Nexa3D, SprintRay, Formlabs, HeyGears, Merit3D, etc.). The vast majority of these companies are dealing with the dental industry, however, some of the businesses are looking into the scale-up production of plastic components for various applications starting from the molds for Injection Molding (IM) and finishing with rapid prototyping.

Similar to other manufacturing process, VPP is not without its challenges, particularly concerning the fabrication of complex parts that include overhangs and inclined edges, where the use of support structures is necessary to ensure the fidelity and integrity of the manufactured parts during the build process. Optimization of the support structures in VPP is crucial for mitigating manufacturing limitations, e.g. fabrication of bridge-like features, overhangs, and islands, among others [3]. Addressing these challenges is essential for the advancement of automated industrial mass production using VPP technology. This paper aims to contribute to this endeavor by presenting a comprehensive investigation into various optimization techniques for support structures in VPP.

The primary objectives of this research are fourfold: first, to identify the minimum amount of support needed for the part to be attached to the support after the build job is finished (henceforth, Support-Free Factor or SFF), second, to develop a benchmark methodology for optimizing support structures in VPP by regulating the process parameters, third, to examine the impact of support removability (before or after the post-processing step) on the ease of detachment and dimensional accuracy of the finished part, and fourth, to identify the most promising method by analyzing the dimensional accuracy and surface topology of the finished samples.

Parts manufactured with fewer support structures may have better surface quality, as there are fewer areas where support material comes into contact with the part. This can be particularly important for applications where surface finish is critical, such as in aerospace or medical industries. Minimizing the support can also lead to material savings, making the manufacturing process more cost-effective, especially for expensive or specialized materials. However, the lack of support can also lead to geometrical errors due to excess material being trapped within the support structure. This paper addresses the exploration of the Support-Free Factor (SFF) for achieving finished geometries with minimal dimensional errors and surface scarring. The term '*scar*' in this investigation is represented as the valleys and/or the peaks, that are formed after the support structure was removed from the part.

By introducing the adaptive slicing method one can control and optimize the Process Parameters (PP) and thickness for different cross-sectional layers of the STL file. It can reduce the time of the build job and tailor operational valuables according to design needs [4].

Jingchao Jiang et.al (2018) in their review on support structure optimization in AM identified the gap in the literature review concerning the "…comprehensive method that can largely reduce the support material while keeping the mechanical strength and good surface finish quality…" [5]. Following up on the research in that field, most investigations focus on design optimization of the support structures and assessing the deformation and strength needed to withstand overhangs [6]. Mizuno Y. et.al (2018) found that overhangs can be consistently fabricated using commercial photopolymers at a 30-degree angle relative to the surface [7]. However, not many studies focus on the quality of the produced parts themselves.

The insights gained from this study hold significant promise for advancing the optimization of support structures in VPP, thus filling the gap in the research field, and paving the way for enhanced efficiency, quality, durability, and scalability in VPP AM.

Nomenclature

AM	Additive Manufacturing
VPP	Vat Photopolymerization
SFF	Support-Free Factor PP – Process Parameters
IM	Injection Molding
STL	Standard Tessellation Language
CAD	Computer Aided Design
IPA	Isopropyl Alcohol

2. Materials and methods

The flowchart represented in Figure 1 is provided to assist the reader in understanding the workflow outlined in the paper.

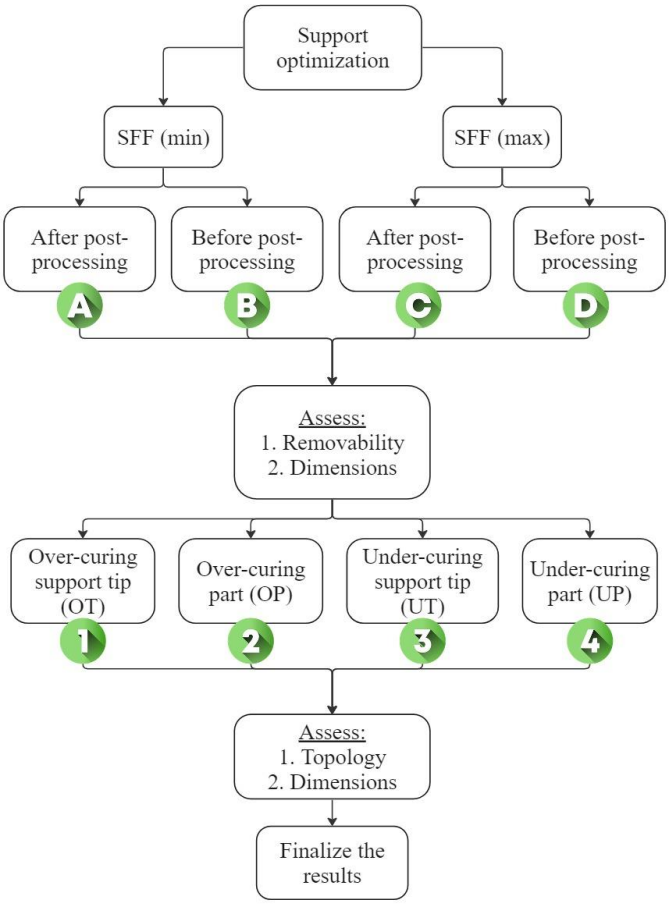


Fig. 1. Workflow of the paper.

The study is divided into two sections, where the first one is dedicated to identifying the optimal parameters for removability and dimensions of the samples. The second study

is delving into the exploration of four methods of slicing adaptation, that affect the final surface topology and dimensional accuracy of the parts.

2.1. Geometry

To achieve the objectives, experiments were conducted focusing on the worst-case scenario: a simple bar geometry positioned parallel to the build surface (0°) with support attached to its underside. Figure 2 represents the CAD model of the part with the support designed using the SolidWorks application. Dimensions of the geometry were adjusted to the pixel size of the setup, where 1 pixel is equal to 7.54 µm in the XY plane.

The support structure was designed using the Lychee slicing software by Mango3D. Support cone settings were kept constant throughout the experiments and were fixed at 0.23 mm for the tip diameter, 1 mm for the tip length and 0.60 mm for the base diameter. Maintaining a minimum resolution of 50 µm in the XY plane for the smallest support features, consistent with the capabilities of commercial VPP systems, thus ensuring replicability. The percentages of pixels of 2D slices were calculated using MathWorks, Inc. MATLAB software.

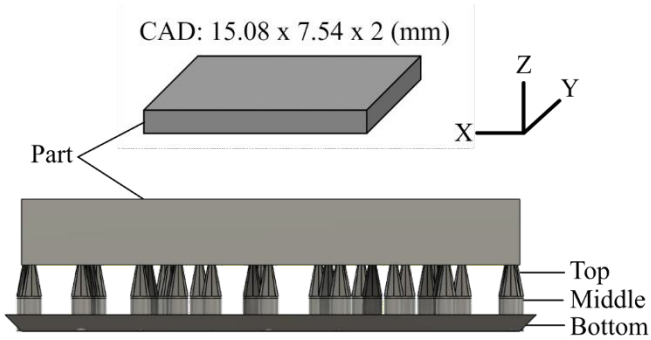


Fig. 2. CAD model of the test geometry.

2.2. Additive manufacturing of photopolymer parts

The experiments were conducted using the open-architecture ultra-high resolution VPP system, developed by the AM group at Technical University of Denmark (DTU) [8], employing commercial photopolymer resin (FTD, Industrial Blend Unpigmented, ρ = 1.05 g/cm³) to validate the findings. PP had been adjusted throughout the experimental work; the summary is shown in Table 1.

Table 1. Process parameters of an open architecture VPP setup.

Process parameters	Value
Layer height, H_l	10 - 50 µm
Burn exposure, E_b	800 ms
Burn irradiance, I_b	13.091 mW/cm^2
Exposure time, E	500 ms
Irradiance, $I_{min} – I_{max}$	9.141 – 19.899 mW/cm^2
Repositioning delay, R	500 ms
Feed rate, v	300 mm/min

Manufactured samples were post-processed using the isopropyl alcohol (IPA) solvent together with an ultrasonic bath at 30°C for 5 minutes with 80 kHz frequency. Following that, samples were post-cured using the 405 nm light for 5 minutes without additional heat.

2.3. Analyzing manufactured samples

Olympus OLS 4000 LEXT laser confocal microscope was used to scan the topology of the parts and subsequently MountainSPIP® software was used for detailed analyzation. Roughness was measured using L-filter (Gaussian ISO 16610-61, 0.025 mm) to remove the long-spatial wavelength. For waviness analysis the S-filter (Gaussian ISO 16610-61, 0.025 mm) was applied to limit the short-spatial wavelength. A digital caliper from Holex with scale divisions 0.01 mm was used to measure the dimensions of the parts.

3. Results and discussion

3.1. Support-Free Factor

By generating geometries with different numbers of the support pillars, the experiment was conducted to identify the minimum amount of support structure needed for the part to be attached to it after the build job is finished. SFF was introduced in this study, and it is expressed in Equation 1:

$$SFF = \frac{A_{unsupported}}{A_{supported}} \quad (1)$$

This equation can subsequently be converted into the relation of the percentage of projected pixels of the supported area and the percentage of pixels for the area of the full part:

$$SFF = \frac{\%_{pixels,\ part} - \%_{pixels,\ support}}{\%_{pixels,\ support}} \quad (2)$$

Table 2 shows the results obtained from this study.

Table 2. Support-free factor effect on the success of the build jobs.

Name	% pixels, support	SFF	Result
S1	3.97	24.19	Success
S2	3.56	27.08	Success
S3	2.99	32.48	Success
S4	2.21	44.17	Success
S5	1.51	65.19	Success
S6	1.34	73.92	Failed
S7	1.16	85.27	Failed
S8	0.81	122.78	Failed

It was observed that with SFF above 65.19 parts were detaching from the support structure, thus the samples failed. The parts that were manufactured with support equal to 1.25% of the object succeeded with high consistency. After

identifying the minimum percentage of the support for the part to be successfully manufactured, further investigation continued with optimization of the support structure specifically for the cases of the SFF_{min} = 65.19 and SFF_{max} = 24.19.

3.2. Optimization of support structure

During the first investigation, four methods of slicing adaptation were studied for parts with SFF_{min}. A graphical representation of each method is depicted in Figure 3.

It has been observed that the dimensional accuracy in the Z-axis of the samples did not correspond to the dimensions of the CAD model. The increase in height of the samples was fixed at approximately Δ = 0.4 mm difference, representing a dimensional error of 20%. Such error was observed due to the photopolymer being trapped between the pillars of the support structure, thus promoting the over-curing of the material.

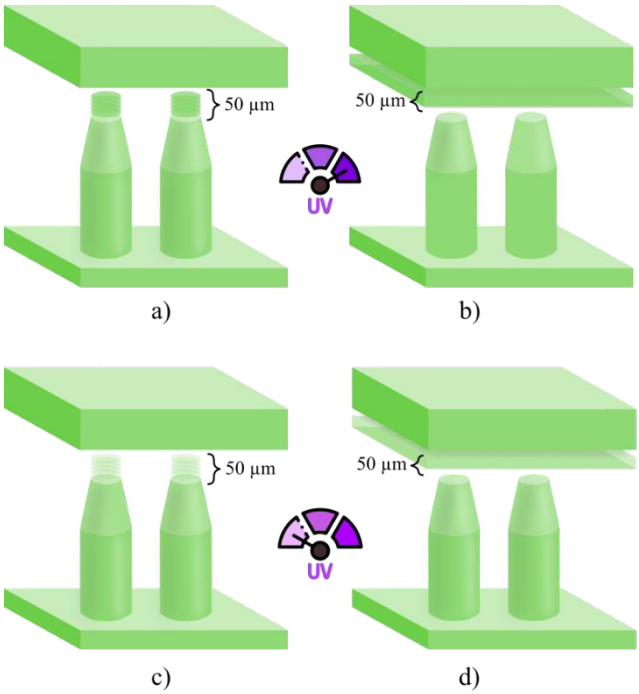


Fig. 3. a) over-exposure of 5 layers (10 µm) within tip of the support; b) over-exposure of 1 layer (50 µm) within part; c) under-exposure of 5 layers (10 µm) within tip of the support; d) under-exposure of 1 layer (50 µm) within part.

To validate this hypothesis, a targeted test was conducted focusing on the support pillars to track the original height of the support structure, as depicted in Figure 4a. Subsequently, five layers with a thickness of 10 µm of the actual part were added atop the support using an irradiance value of 9.141 mW/cm², as shown in Figure 4b. It could already be seen how the layers of the part penetrate through the support. Increasing the UV projection irradiance within the part to 19.899 mW/cm², as depicted in Figure 4c, demonstrated the emergence of photopolymer buildup within the support pillar, confirming the entrapment phenomenon.

In Figure 4d, adjusting the layer thickness of the part itself from 10 µm to 50 µm effectively mitigated the risk of excessive UV light penetration within the support structure, hence leading to a better dimensional accuracy in the Z-axis.

A layer thickness of 50 µm was selected to optimize both structural integrity and interlayer adhesion within the part, thereby ensuring its overall quality. After the optimization of the PP, the error within the height of the part was assumed to be <10% (200 µm).

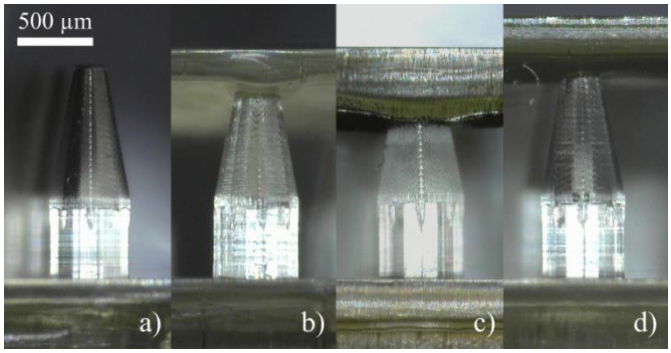


Fig. 4. a) support pillar, no part; b) support pillar and five 10 µm layers of part; c) support pillar and five 10 µm layers of part, over-exposed; d) support pillar and two 50 µm layers of part.

Removability of the part from the support depends on factors such as timing of removal (before or after the post-processing), technique (pilling, twisting etc.) and SFF. To evaluate the removability, the quantitative Design of Experiment (DoE) was conducted, as seen in Table 3. Removal techniques were not included in this study.

The ranking for removability was set from 1 to 5 points, where 1 – very easy to remove, 2 – easy to remove, 3 – moderate, 4 – hard to remove, and 5 – very hard to remove. The experimental procedures were conducted by an individual with expertise in AM, with no consideration given to gender or age factors within the study.

Table 3. DoE for analyzing the removability.

	Parameters			
SFF	SFF_{min}		SFF_{max}	
Timing	After post-processing	Before post-processing	After post-processing	Before post-processing
Case name	A	B	C	D
Ranking	3	1	3	1
Dimension Z, [mm]	2.24 ± 0.02	2.21 ± 0.03	2.17 ± 0.01	2.18 ± 0.01

The findings suggest that cases B and D demonstrated the highest ease of removal, attributable to the initial loose contact between the part and support material immediately after manufacturing. On the contrary, cases A and C exhibited moderate difficulty in removal from the support, as the post-processing steps led to the hardening of the support material. Evaluation of Z height showed that cases C and D were closest to the original value of 2 mm, because of increased amount of the support structure. Following the outcome of the study, it was decided to continue the experiments with SFF_{max} and remove the parts from support before the post-processing stage. In the following section, the effects of the four methods from

Figure 3 on removability, dimensional accuracy, and topology of finished parts are discussed.

3.3. Effect of slicing adaptation on dimensional accuracy, and surface topology

During this study the code names for the samples were introduced: OT (over-curing within the tip of the support), OP (over-curing within the part), UT (under-curing within the tip of the support and UP (under-curing within the part). For over-curing I_{max} = 19.899 mW/cm^2 and for under-curing I_{min} = 9.141 mW/cm^2 was applied. The results of the dimensional accuracy of samples in the Z direction can be seen in Figure 5.

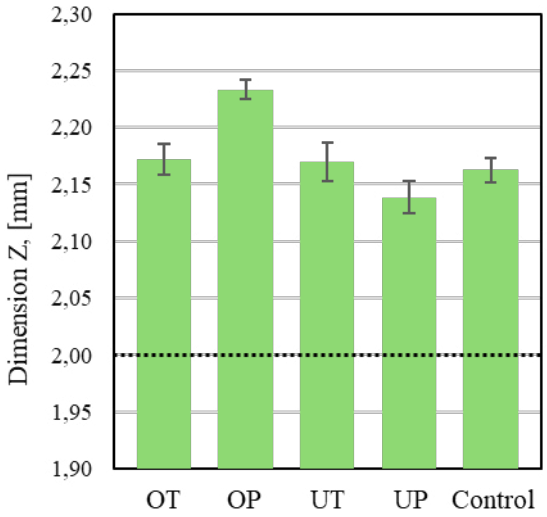


Fig. 5. Dimensions of samples using four methods. Dashed line represents the nominal value of 2 mm.

Samples subjected to both over- and under-exposure of UV light within the support structure tip (OT and UT), alongside control samples, demonstrated similar thickness results, averaging approximately 8.5% higher than the nominal value. On the contrary, samples with layers experiencing under-curing within the part (UP) have exhibited values closest to the nominal, resulting in an error of 6.9%. The worst result was tracked for the over-curing within part (OP), showing an error of approximately 11.7%.

The findings indicate that manipulations with UV intensity within the support structure tip have minimal effect on dimensional changes, with thickness remaining relatively consistent. However, variations in UV exposure within the part layers result in significant deviations from nominal values, with under-curing yielding the closest results and over-curing leading to the largest errors. This phenomenon is explained due to the fact, that when UV intensity is adjusted within the layers of the part itself, it directly impacts the curing process of the part, therefore directly influencing the thickness. However, when UV light is manipulated within the tip of the support structure, it primarily affects the curing of the support material itself rather than the adjoining part.

When looking at the production of parts for industrial applications, not only the dimensions but also the surface quality plays a very big role. In this study roughness of the same samples is shown in Figure 6. Comparing these results to the control samples, which were manufactured without any manipulation of UV curing, it's evident that adaptive slicing methods have a notable impact on surface roughness. The control samples exhibit higher roughness (Sq = 3.960 µm) compared to samples manufactured with adaptive slicing methods.

Considering the roughness values, the samples manufactured with low UV irradiance within the part itself demonstrated the lowest surface roughness among all tested components. Furthermore, the standard deviation for this method (0.219 µm) is considerably lower than that of other methods, indicating greater consistency in surface quality.

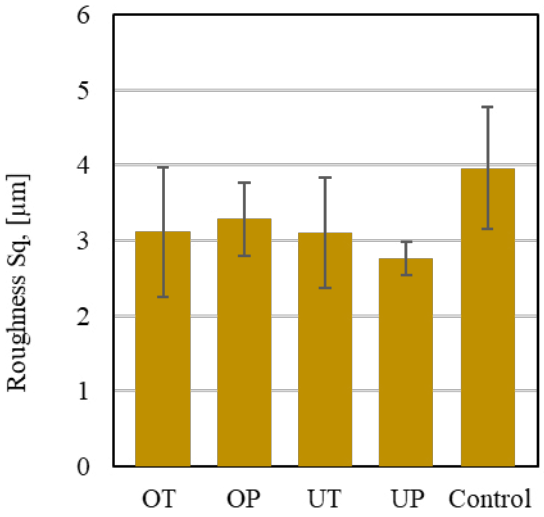


Fig. 6. Surface roughness analysis of samples using four methods.

4. Conclusion

This study delved into the critical relationship between support structures and final part quality in VPP AM. Through systematic experimentation, the minimum required support for ensuring part attachment post-build was determined to be SFF_{min} = 65.19. It was observed that a lower amount of support diminished dimensional accuracy in the Z-axis, resulting in errors of up to 10.5%. However, the impact of support removability on the ease of detachment, topology and dimensional accuracy of the finished part showed, that parts manufactured with a high amount of support (SFF_{max} = 24.19) removed before the post-processing reduced the error in thickness to 8.67%, as well as resulted in the smoother surface of the samples.

Moreover, the investigation into adaptive slicing techniques revealed promising avenues for optimizing part thickness. While altering mask projection irradiance directly impacted dimensional accuracy, PP optimization within the support tip showed limited influence on part thickness, primarily affecting support layers. Consequently, the method utilizing low UV irradiance within the part itself emerged as the most effective, yielding thickness errors as low as 6.9% and delivering a smoother surface finish.

5. Future work

Continuing this research, further exploration into PP optimization for the preferred manufacturing method is warranted to minimize thickness errors to less than 5%. A deeper investigation of the surface topology of parts produced with SFF_{min} in comparison to SFF_{max} could be beneficial for the qualitative analysis. Additionally, investigating various techniques for assessing part removability from support structures, such as pulling or twisting, could enhance the understanding of automated production possibilities in photopolymer part manufacturing, as well as improve the surface roughness of the parts. Evaluation of material that can be saved by applying SFF, could help to understand the scale of the sustainable future of this approach. These future endeavors aim to advance the efficiency and reliability of VPP AM, ultimately facilitating the transition toward automated industrial-scale production.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used OpenAI "ChatGPT" Version 3.5 to find synonyms for words and clarify the correctness of language for scientific writing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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